Nonlinear Dyn (2025) 113:25801–25819
https://doi.org/10.1007/s11071-025-11439-y

RESEARCH

Experimental and mathematical simulation of multi-mode VIV of long span bridge

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Received: 18 March 2025 / Accepted: 28 May 2025 / Published online: 6 July 2025

Abstract Vortex induced vibration (VIV) of long span bridge is becoming an urgent problem in modern society. Because of the dense modal distribution in frequency domain, VIV of different modes can be excited in sequence within a narrow range of wind speed, which is well known as multi-mode VIV. However, such the multi-mode VIV phenomenon has not been well addressed theoretically. In this study, experimental and mathematical simulation is conducted for understanding and modelling multi-mode VIV, using a multi-span elastically supported beam. One of the interesting feature of multi-mode VIV observed in experiment is that, VIV of different modes are mutually exclusive, even though the lock-in ranges can be overlapped. The dual-oscillator model is firstly applied for the mathematical VIV simulation of a single mode. Then it is further extended to explore the multi-mode VIV of long span bridge by embedding the model into the governing function of a long span bridge. The proposed model shows good feasibility in simulating both the single- and multi-mode VIV, as the model can capture the most critical feature observed from the experiment. By examining the governing function of mathematic model, it is found that the Van Der Pol type nonlinear damping is responsible for the mutual exclusive feature between VIV of neighboring modes. The incremental harmonic balance method is adopted for detailed investigation of the multi-DoF nonlinear dynamic system. Two solution branches are identified, providing more comprehensive understanding of the multi-mode VIV characteristics of long span bridge.

Keywords Multi-mode VIV · Long span bridge · Mathematic model · Nonlinear dynamics · Incremental harmonic balance method

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1 Introduction

Vortex induced vibration (VIV) is one of the most critical problem for slender or long span structures [1–3]. Although VIV is known as limited amplitude vibration, it is one of the major sources inducing fatigue problem or even the threat of safety for slender structures [4, 5]. VIV happens when the vortex shedding frequency approaches the natural frequency of structure, so that the resonance can be induced. One recent well-known VIV event is observed on Humen bridge at May 5th, 2020, Guangzhou, China [6]. The maximum vibration of Humen bridge during the VIV reached 0.3m, and the local government had to temporarily close the ground transportation through this bridge and the shipping channel under the bridge.

VIV has been extensively studied by researchers in the past decades [7–9]. Studies found that one critical problem of VIV is that, structure response during VIV is not a function of incident wind only, it is also a function of structure vibration itself. In other words, VIV can induce the fluid–structure interaction (FSI) [10–13]. A typical phenomenon induced by such interaction is known as lock-in, which means the vortex shedding frequency is locked at the structure's natural frequency within a wind speed range. In this case, the resonance of the structure can be induced with experiencing large amplitude vibration [14, 15]. Many studies have, thus, investigated and modeled FSI [16, 17]. They can roughly be classified into two approaches from the view point of how to interpret FSI into dynamic functions. One is the self-excitation force model [18, 19], and the other is the wake oscillator model [20, 21].

For self-excitation force model, the aeroelastic damping force and stiffness force, which are function of structure velocity and displacement, are inserted into the structure's governing function to describe FSI [22, 23]. Ehsan and Scanlan borrowed from the van der Pol oscillator form, and the self-excitation force is described as the structural velocity multiplied by the square of structural displacement [24]. Zhu et al. proposed that the self-excitation force is associated with the cubic of structural velocity [25, 26]. Song et al. [27] proposed that the self-excited force could be expressed by a third-order Taylor expansion with respect to velocity and displacement response. Hui et al. [28] proposed an amplitude-adaptive self-excited force model, and the model parameters do not need to be adjusted with wind speed. Xu et al. [29] examined the capabilities of some commonly used single-degree-of-freedom (SDOF) VIV models for bridge decks through wind tunnel experiment, and claimed that the determination of aerodynamic damping force is very important for the estimation of self-excited force. Currently, the self-excitation force model has been widely used in wind engineering field, but this kind of models are basically established based on the curve fitting method, with no clear underlying physics.

Wake oscillator model shows advantage on physical interpretation of VIV. This type of model originally interprets VIV from the mathematic point of view, as the lock-in phenomenon can be treated as a typical feature of nonlinear dynamic system [30]. Hartlen and Currie's model [31] and Iwan and Blevins's model [32] show self-excited and self-limited natures with referring to Van der Pol function, which could predict the maximum structural amplitude. However, these models encountered significant limitation in accurately simulating response of the entire VIV process, i.e. the VIV wind speed range obtained from them may narrower than the target one [33]. Meanwhile, those models mostly focused on the mathematics with neglecting the mechanism of VIV. Tamura's model overcame the above problem by deriving a wake-oscillator model for the circular cylinder with clear physical meaning of the involved parameters. This model offers more physical representation of the wake's sway motion, and is able to predict the entire vortex vibration process of cylindrical structures accurately, which led to the widely application [33, 34]. Although Tamura's model can successfully simulate the lock-in phenomenon, it is limited in modelling the VIV of circular and square section cylinder [35]. More recently, Hui et al. [36] proposed the dual-oscillator model to tackle the VIV of rectangular section model, which can well consider the reattachment of separated flow from the leading corner. Introduction of the Dual-oscillator model is given in Sect. 3.1.

According to the above introduction, it can be known that modeling of VIV has been extensively studied for decades [37, 38]. Even though, it should be noticed that, all those models are basically proposed for simulating single mode VIV [39–41]. However, for real structures, especially long span bridges, cables and risers, their modal frequencies are always densely distributed [42–44]. In this case, it can be expected that, the VIV wind speed ranges of two neighboring modes may overlap each other partially. Several interesting phenomena have been observed from experiment and field measurement for long span bridges, such as VIV occurs in sequence for different modes, and the maximum VIV amplitudes are proximately constant for different modes [45–47]. Now, the problem becomes that comprehensive modelling research for multi-mode VIV is still insufficient [48–51].

In this study, the multi-mode VIV is firstly investigated through wind tunnel experiment, with the modal parameters being identified before wind tunnel test. Then the dual-oscillator is further developed for simulating the multi-mode VIV, by considering two neighboring modes with overlapped lock-in ranges. At last, the mathematical model of the multi-mode VIV is investigated to explain the mechanical reason of the observed characteristics of multi-mode VIV.

2 Wind-tunnel experiment

2.1 Model information

An 8.1 m long elastic continuous beam was adopted in the wind tunnel experiment, and it is not used for simulating any real bridge. 17 elastic supports were uniformly distributed along the beam, with two ends hinge supported, as shown in Fig. 1. The stiffness of the 17 supports are the same Ki = 120N/m (i = 1, 2, …, 17). The cross section of the beam is a 6:1 rectangular section, with 0.04 m depth (D), 0.24 m breadth (B). The stiffness of the beam itself is provided by the steel plate in the girder (Fig. 1), which has the width of 80 mm and thickness 4 mm, resulting EI = 84 N m^2. The mass is 3.0 kg/m along the spanwise direction. The stiffness of the beam itself is provided by a steel plate as shown in Fig. 1b. The Model's information is shown in Table 1 [52–54]. To accurately track the mode shapes of the model, 16 accelerometers are mounted on the beam, and Fig. 2 shows the mode shapes of the first 8 modes of the experimental model, and Table 2 shows the corresponding modal frequencies and damping ratios.

Mode shape of the jth mode follows Eq. (1):

$$\phi_j(x) = \sin\left(\frac{j\pi}{L}x\right). \tag{1}$$

2.2 Experiment condition

The wind tunnel test was conducted in HD-2 wind tunnel at Hunan University, with the cross-section of 8.5 m × 2.0 m. Wind speed of the wind tunnel can be adjusted from 0 to 16 m/s. Uniform flow was used for testing the VIV of target beam in the experiment. The turbulence intensity of the inflow is 1%. 9 accelerometers were used to measure the vibration response of the aero-elastic model, and they were placed at 3L/24, 4L/24, 6L/24, 9L/24, 12L/24, 15L/24, 18L/24, 20L/24, and 21L/24. More details about the experiment can be found in [47]. Figure 3 shows the wind tunnel

Table 1 Model information

Length (m)	Breadth (m)	Depth (m)	EI (N·m^2)	Ki (N/m)
8.1	0.24	0.04	84	120

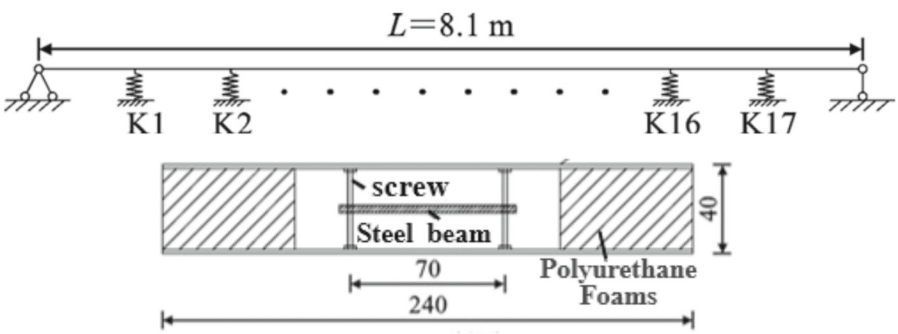
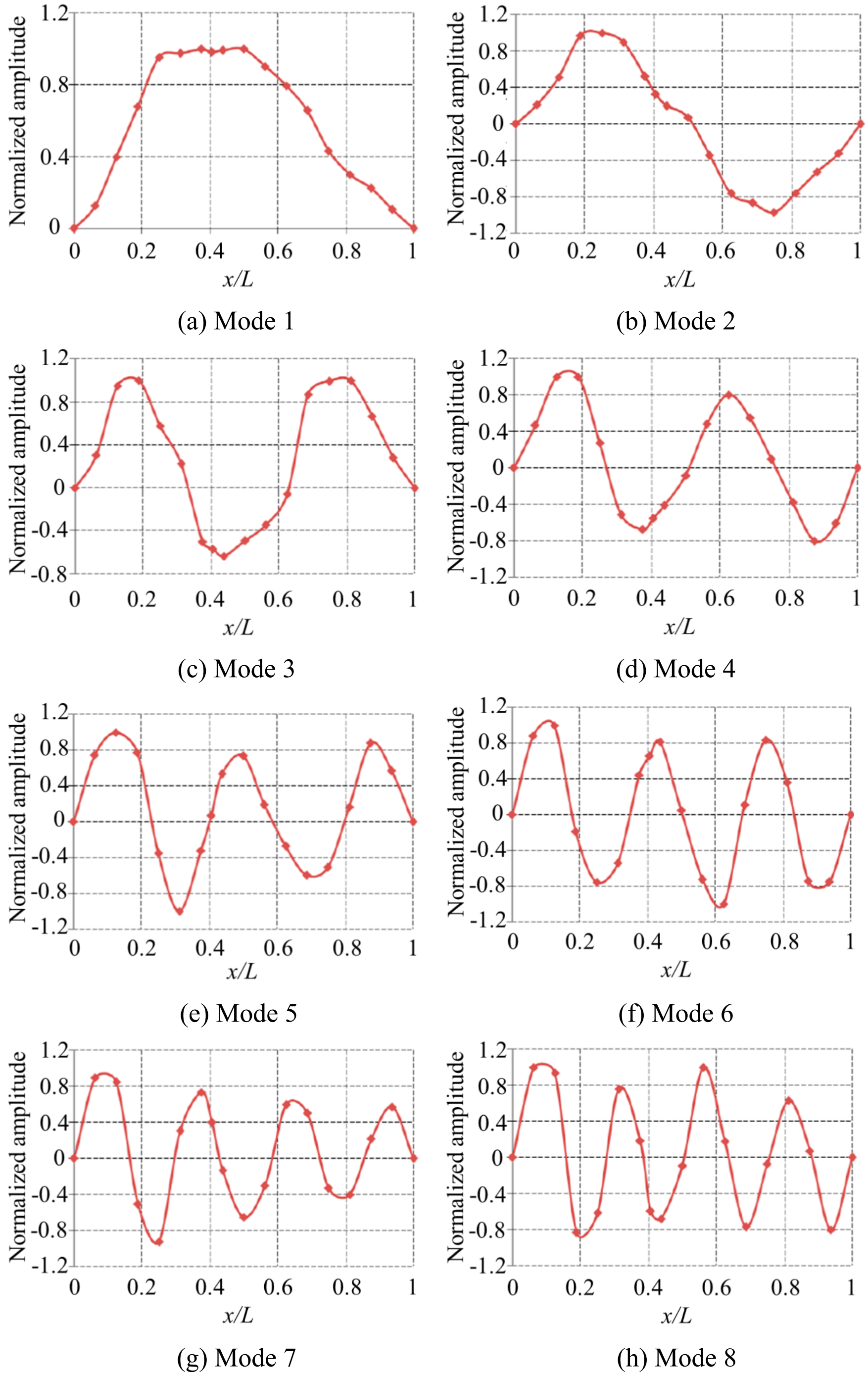


Fig. 1 Design diagram of elastically multi-supported aeroelastic beam

Fig. 2 Measured mode shapes of the beam



experiment model of the aeroelastic beam. Stiffness of the elastic supports are provided by the cantilever steel bars as shown in Fig. 3.

2.3 Experimental results

Figure 4 shows the variation of the vibration amplitudes of each mode with the increase of wind speed.

Table 3 shows the wind speed range of VIV of each mode and their corresponding maximum response amplitudes.

According to the results shown in Fig. 3 and Table 3, it can be clearly checked that VIV of different modes are excited sequentially with the increase of incident wind speed. The turbulence intensity gradually decreases and becomes stable, when wind speed increases. Meanwhile, it can also be checked that there is no overlap for the wind speed range of VIV of adjacent modes. Sudden increase of the subsequent mode's response can be observed immediately after the VIV of previous mode vanished. In other word, for each incident wind speed, only VIV of one mode can be excited, no matter how close of the modal frequency of the two neighboring modes is. This may be because that when the VIV of one mode is induced, the vortex shedding frequency is locked at certain value, and VIV of the following mode can only be induced after the current lock-in is broken. The maximum VIV amplitude can also be checked gradually approaching stable, and this is because of the turbulence intensity of the incident wind becomes stable in high speed region [55].

In order to mathematically simulate such multi-mode VIV phenomenon, a feasible model is required. In the following of this study, the dual-oscillator model will be adopted and further extended for this purpose.

Table 2 Mode information of target beam

Vibration mode	Frequency (Hz)	Damping ratio (%)
Mode 1	2.73	0.39
Mode 2	3.61	0.55
Mode 3	4.49	0.63
Mode 4	5.18	0.70
Mode 5	5.86	0.61
Mode 6	6.74	0.63
Mode 7	7.86	0.60
Mode 8	9.18	0.60

3 Simulation of single mode VIV

3.1 Dual-oscillator model

Dual-oscillator model was proposed by Hui et al. [36] for simulating VIV of rectangular section cylinder with aspect ratio 4:1 (B:D). The model can mathematically simulate the periodic variation of both main vortices and wake vortices, which are generated from the leading edges and trailing edges, respectively, as shown in Fig. 5.

Governing function of model is as follows,

$$\ddot{\alpha}_1(t) - 2\omega_v\zeta_1\left[1 - \frac{4f_{m,1}^2}{\hat{C}_{L1}^2}\alpha_1^2(t)\right]\dot{\alpha}_1(t) + \omega_v^2\left[\alpha_1(t) - \frac{\dot{y}(t)}{U}\right] = \frac{\ddot{y}(t)}{\bar{l}_1}, \tag{2a}$$

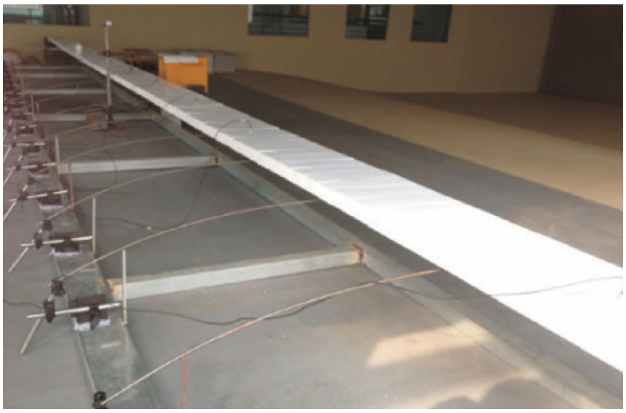


Fig. 3 Multi-supported aeroelastic beam

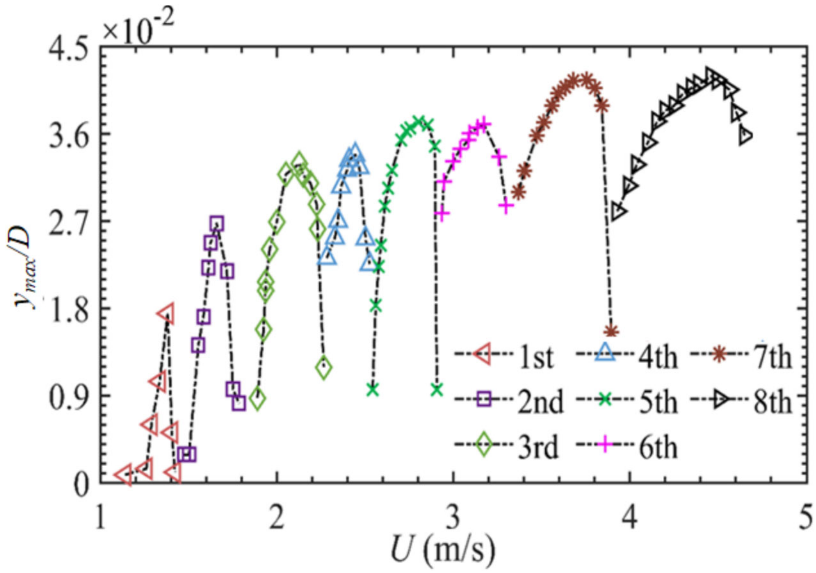


Fig. 4 Response amplitude of each mode varying with wind speed

Table 3 Experimental results

Mode	Mean wind speed $\overline{U}$ (m/s)	Turbulence intensity I_t (%)	$y_{\max}/D$
1	1.14–1.42	12.70–10.60	0.017
2	1.48–1.79	8.77–5.89	0.027
3	1.89–2.26	5.61–5.10	0.033
4	2.24–2.53	3.53–3.19	0.034
5	2.43–2.91	3.34–2.48	0.037
6	2.91–3.30	2.24–1.96	0.037
7	3.37–3.90	1.89–1.47	0.042
8	3.92–4.65	1.39–1.10	0.042
9	4.72–5.44	1.13–1.00	0.042

Turbulence intensity is defined as Std(U)/$\overline{U}$

$$\ddot{\alpha}_2(t) - 2\omega_v\overline{\zeta}_2\gamma\left[1 - \frac{4f_{m,2}^2}{\gamma^2\hat{C}_{L2,v}^2}\alpha_2^2(t)\right]\dot{\alpha}_2(t) + \omega_v^2\left[\alpha_2(t) - \frac{\dot{y}(t)}{U}\right] = \eta\omega_v\dot{\alpha}_1(t) + \frac{\ddot{y}(t)}{\overline{l}_2}, \tag{2b}$$

$$m_s\ddot{y}(t) + 2\omega_s m_s\zeta_s\dot{y}(t) + m_s\omega_s^2 y(t) = \frac{1}{2}\rho U^2 DLC_L(t), \tag{2c}$$

$$C_L(t) = C_{L1}(t) + C_{L2}(t) = f_{m,1}\left[\alpha_1(t) - \frac{\dot{y}(t)}{U}\right] + f_{m,2}\left[\alpha_2(t) - \frac{\dot{y}(t)}{U}\right], \tag{2d}$$

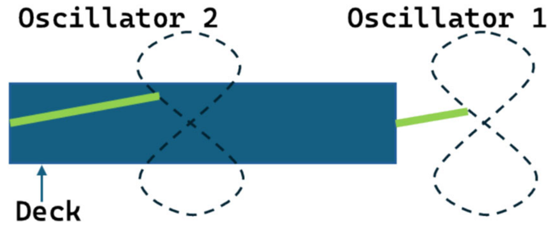


Fig. 5 Dual-oscillator

$$\hat{C}_L = \hat{C}_{L1} + \hat{C}_{L2},\ \hat{C}_{L2} = \hat{C}_{L2,v}/\gamma,\ \hat{C}_L = \hat{C}_{L2,v}/\lambda, \tag{2e}$$

where, α_1 and α_2 are the swaying angles of Oscillator 1 and 2, respectively; y is the displacement of the vibrating cylinder; D is the depth of cross section; $\hat{C}_{L1}$,

$\hat{C}_{L2}$ are the amplitude of the lift force coefficients $C_{L1}(t)$ and $C_{L2}(t)$, respectively; $\hat{C}_{L2,v}$ is amplitude of the lift force coefficients induced by main vortex only [36]; ζ_i, (i = 1, 2, and $\zeta_2 = \overline{\zeta}_2\gamma$) is the damping ratio of Oscillator i; $f_{m,i}$ is the Magnus effect coefficient of Oscillator i; $\bar{l}_i$ is the half mean length of Oscillator i; ω_v is the natural frequency of Oscillators 1 and 2, which is also known as the vortex shedding frequency; γ, η are the weighing coefficients; m_s, ω_s, and ζ_s is the mass, circular frequency, and structural damping ratio of the cylinder, respectively.

In Eq. (2), the first and second equations (Eqs. (2a–2b)) are the governing functions of the Oscillators 1 and 2. To ensure that Oscillator 2 vibrates in phase with Oscillator 1, the affection of Oscillator 1 to Oscillator 2 is proportional to $\dot{\alpha}_1(t)$. Equation (2c) is the governing function of the cylinder (deck). It can be checked that the cylinder is a linear dynamic structure, and it is excited by the two nonlinear oscillators simultaneously. The two oscillators are self-excited following the Van Der Pol function, which are also affected by the oscillation of cylinder as shown on the right-hand side of equation. Details of the model can be check from Hui et al. [36].

The above model is proposed for 4:1 rectangular cylinder, for which the flow has reattachment on the side face. Similar feature also fits the 6:1 rectangular cross section [56]. Therefore, the model is applicable for the 6:1 rectangular cylinder. However, this model is proposed for the VIV of a cylinder section, and only VIV of one specified mode can be simulated. In this study, however, a long span beam is selected as simulation target, which exhibits sequential VIV of several different modes.

Thus, the model needs to be further improved. The response in Eq. (2) is defined in distributed form, as

$$y(z,t) = \phi(z)\cdot(D\cdot q(t)), \tag{3a}$$

$$\alpha_1(z,t) = \phi(z)\cdot\beta_1(t), \tag{3b}$$

$$\alpha_2(z,t) = \phi(z)\cdot\beta_2(t), \tag{3c}$$

where, $\phi(z)$ is the mode shape function, z is the span-wise position, $\beta_1(t)$, $\beta_2(t)$, $D{\cdot}q(t)$ are the generalized coordinates of oscillators and structure, respectively; $q(t)$ is the normalized modal displacement with respect to D.

By further introducing dimensionless variables such as time ($\tau = \omega_s t$), and reduced wind speed ($U_R = 2\pi U/(\omega_s D)$), the non-dimensional governing functions of the model for single mode (mode j, $\omega_s = \omega_j$) can be presented as follows,

$$\begin{aligned}&\beta''_{1,j}(\tau) - 2\zeta_1 U_{R,j}\cdot St\cdot\left[1-\frac{\int_0^L\phi_j^4(z)\mathrm{d}z}{\int_0^L\phi_j^2(z)\mathrm{d}z}\cdot\frac{4f_{m,1}^2}{\hat{C}_{L1}^2}\beta_{1,j}^2(\tau)\right]\\&\quad\beta'_{1,j}(\tau) + U_{R,j}^2\cdot St^2\cdot\beta_{1,j}(\tau)\\&\quad=\frac{D}{\bar{l}_1}q''_j(\tau) + 2\pi U_{R,j}\cdot St^2\cdot q_j{\prime}(\tau)\end{aligned} \tag{4a}$$

$$\begin{aligned}&\beta''_{2,j}(\tau) - 2\gamma\overline{\zeta}_2 U_{R,j}\cdot St\cdot\left[1-\frac{\int_0^L\phi_j^4(z)\mathrm{d}z}{\int_0^L\phi_j^2(z)\mathrm{d}z}\cdot\frac{4f_{m,2}^2}{\hat{C}_{L2,v}^2}\beta_{2,j}^2(\tau)\right]\\&\quad\beta'_{2,j}(\tau) + U_{R,j}^2\cdot St^2\cdot\beta_{2,j}(\tau) = \frac{D}{\bar{l}_1}q''_j(\tau) + 2\pi U_{R,j}\\&\quad\cdot St^2\cdot q_j{\prime}(\tau) + \eta U_{R,j}\cdot St\cdot\beta_{1,j}{\prime}(\tau)\end{aligned} \tag{4b}$$

$$\begin{aligned}&q''_j(\tau) + \left[2\zeta_{s,j} + \frac{\rho D^2 L}{2m_j}\cdot\left(\frac{U_{R,j}}{2\pi}\right)\cdot(f_{m,1}+f_{m,2})\right]q'_j(\tau) + q_j(\tau)\\&\quad=\frac{\rho D^2 L}{2m_j}\cdot\left(\frac{U_{R,j}}{2\pi}\right)^2\cdot\left[f_{m,1}\cdot\beta_{1,j}(\tau) + f_{m,2}\cdot\beta_{2,j}(\tau)\right]\end{aligned} \tag{4c}$$

where $St = \omega_v D/(2\pi U)$ is Strouhal number, $m_j = \int_0^L \phi_j(z)m_s(z)\phi_j(z)\mathrm{d}z$.

3.2 Parameter identification and validation based on single mode

Before the simulation of multi-mode VIV, the VIV process of each single mode should be simulated. According to Eq. (2), several parameters need to be decided in prior, including: $\hat{C}_L$, $\hat{C}_{Lj}$, $\hat{C}_{L2,v}$, ζ_j, $f_{m,j}$, $\bar{l}_i$ (i = 1, 2), ω_v, γ, η, m_j, ω_j and $\zeta_{s,j}$.

By examining the response shown in Fig. 3 and Table 2, it can be found that for the cases of VIV of lower modes (Mode 1–6), the response amplitudes are slightly lower than higher modes (Mode 7–8). This is because for VIV of those lower modes, the incident wind speed is low with relatively high turbulence, and a large turbulence intensity can suppress the response of VIV [55, 57].

Therefore, VIV of the 7[th] mode in the above experiment is selected as the target, and some parameters can be determined accordingly, with m_7 = 25.11 kg, $\omega_{s,7}$ = 49.39 rad/s, and $\zeta_{s,7}$ = 0.60%, St = 0.093. Besides, $\hat{C}_L$ and $\bar{l}_i$ (i = 1, 2) are determined with referring to the study carried out by Li et al. [56], which is also a 6:1 rectangular section. In that experiment, the average length of main vortex decreased significantly from 4.6D to 3.5D when the inflow condition slightly varies from uniform flow to turbulent flow (1.5% turbulence intensity). In this study, as the turbulence intensity of inflow is 1%, the average length of main vortex ($2\bar{l}_2$) is thus assumed as 3.8D. The average length of wake vortex ($\bar{l}_1$) keeps constant around 1.0D for various cases in that study. Consequently, to model the VIV $\bar{l}_1 = D, \bar{l}_2 = 1.9D$ are adopted. Moreover, $\hat{C}_L = 0.5$ is used according to that study. In addition, λ = 0.25 is used with referring to the Matsumoto et al.'s results [58] obtained for rectangular section. Then $\hat{C}_{L2,v}$ can be fixed by using Eq. (2e)

$$\hat{C}_{L2,v} = 0.25 \cdot \hat{C}_L = 0.125. \tag{5}$$

Meanwhile, according to Hui et al. [36], by setting $\hat{C}_L = 0.5$, the relation of γ and η can be obtained as:

$$\eta = 0.018 \times \gamma^{-3.9}. \tag{6}$$

Finally, $f_{m,1}$ = 2.7, $f_{m,2}$ = 14.5, γ = 0.54, $\hat{C}_{L1} = 0.27$, $\hat{C}_{L2} = 0.23$, ζ_1 = 0.1, $\bar{\zeta}_2 = 0.27$ can be identified by fitting the wind speed-response amplitude curve to the experimental results with accordance to Hui et al. [36]. Figure 6a compares the response amplitude curve of the 7th mode predicted by the proposed model and that obtained from experiment. It is clear that the two results match well.

In order to examine the feasibility of the model, by setting $\zeta_{s,8}$ = 0.60%, $\omega_{s,8}$ = 57.68 rad/s, with all the other parameters unchanged, the response curve of mode 8 can be predicted by the model. The obtained results are also compared with the experimental results, showing good agreement with each other (Fig. 6b), $U_{R,j} = 2\pi U/(\omega_j D)$, although it is slightly narrower in the wind speed range. Such difference may be due to the slight difference between real mode shape and the ideal one used in the simulation or some other experiment uncertainties. These results indicate that, the proposed model is capable of predicting the VIV of each specific mode of the target aeroelastic beam, which provides a solid base for simulation and analysis of multi-mode VIV.

4 Simulation of multi-mode VIV

4.1 Derivation of governing function

To simulate the multi-mode VIV, the response in Eqs. (2) and (3) need to be re-defined,

$$y(z,t) = \sum_{j=n}^{n+i} \phi_j(z) \cdot \big(D \cdot q_j(t)\big), \tag{7a}$$

$$\alpha_1(z,t) = \sum_{j=n}^{n+i} \phi_j(z) \cdot \beta_{1,j}(t), \tag{7b}$$

$$\alpha_2(z,t) = \sum_{j=n}^{n+i} \phi_j(z) \cdot \beta_{2,j}(t), \tag{7c}$$

where, n and i are positive integers. Take Eq. (7) into Eq. (2), one can get,

$$\begin{aligned}&\sum_{j=n}^{n+i} \phi_j(z) \cdot \ddot{\beta}_{1,j}(t) - 2\omega_v \zeta_1 \\ &\quad \left\{1 - \frac{4f_{m,1}^2}{\hat{C}_{L1}^2}\left[\sum_{j=n}^{n+i} \phi_j(z) \cdot \beta_{1,j}(t)\right]^2\right\} \sum_{j=n}^{n+i} \phi_j(z) \cdot \dot{\beta}_{1,j}(t) \\ &\qquad + \omega_v^2 \left[\sum_{j=n}^{n+i} \phi_j(z) \cdot \beta_{1,j}(t) - \frac{D \cdot \sum_{j=n}^{n+i} \phi_j(z) \cdot \dot{q}_j(t)}{U}\right] \\ &\quad = \frac{D \cdot \sum_{j=n}^{n+i} \phi_j(z) \cdot \ddot{q}_j(t)}{\bar{l}_1}\end{aligned} \tag{8a}$$

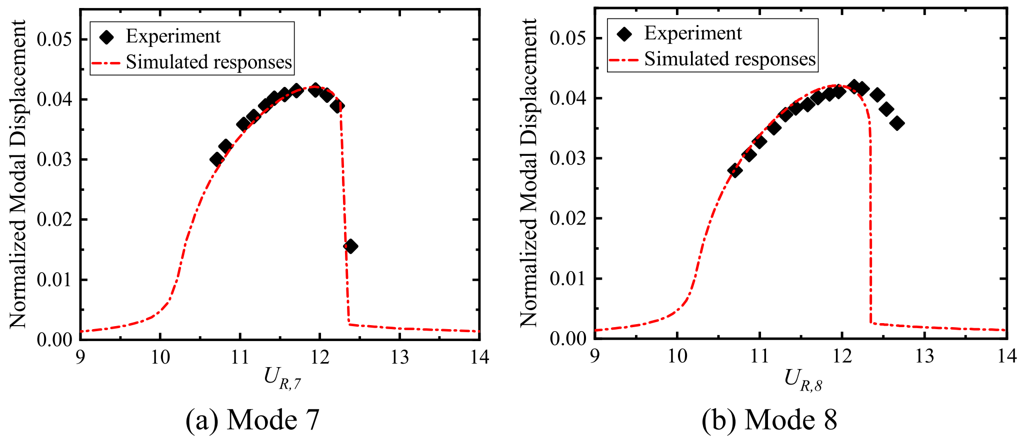


Fig. 6 Modal response amplitude versus reduced wind speed

$$\sum_{j=n}^{n+i}\phi_j(z)\cdot\ddot{\beta}_{2,j}(t)-2\omega_v\overline{\zeta}_2\gamma\left\{1-\frac{4f_{m,2}^2}{\hat{C}_{L2,v}^2}\left[\sum_{j=n}^{n+i}\phi_j(z)\cdot\beta_{2,j}(t)\right]^2\right\}\sum_{j=n}^{n+i}\phi_j(z)\cdot\dot{\beta}_{1,j}(t)+\omega_v^2\left[\sum_{j=n}^{n+i}\phi_j(z)\cdot\beta_{2,j}(t)-\frac{D\cdot\sum_{j=n}^{n+i}\phi_j(z)\cdot\dot{q}_j(t)}{U}\right]=\eta\omega_v\sum_{j=n}^{n+i}\phi_j(z)\cdot\dot{\beta}_{1,j}(t)+\frac{D\cdot\sum_{j=n}^{n+i}\phi_j(z)\cdot\ddot{q}_j(t)}{\overline{l}_2} \tag{8b}$$

$$\sum_{j=n}^{n+i}m(z)\phi_j(z)\cdot\ddot{q}_j(t)+2\omega_n\zeta_s\sum_{j=n}^{n+i}m(z)\phi_j(z)\cdot\dot{q}_j(t)+\omega_n^2\sum_{j=n}^{n+i}m(z)\phi_j(z)\cdot q_j(t)=\frac{1}{2}\rho U^2L\cdot C_L(t) \tag{8c}$$

$$C_L(t)=f_{m,1}\left[\sum_{j=n}^{n+i}\phi_j(z)\cdot\beta_{1,j}(t)-\frac{D\cdot\sum_{j=n}^{n+i}\phi_j(z)\cdot\dot{q}_j(t)}{U}\right]+f_{m,2}\left[\sum_{j=n}^{n+i}\phi_j(z)\cdot\beta_{2,j}(t)-\frac{D\cdot\sum_{j=n}^{n+i}\phi_j(z)\cdot\dot{q}_j(t)}{U}\right] \tag{8d}$$

Take mode 7 and 8 as example to verify the above functions, with n = 7, i = 1. After non-dimensionalization and mode discretization, the following functions can be obtained:

$$\beta_{1,7}''(\tau)-2\zeta_1 v\cdot\left\{1-\frac{4f_{m,1}^2}{\hat{C}_{L1}^2}\cdot\left[e_{p7}\beta_{1,7}^2(\tau)+e_{p78}\beta_{1,8}^2(\tau)\right]\right\}\beta_{1,7}'(\tau)+v^2\cdot\beta_{1,7}(\tau)=\frac{D}{\overline{l}_1}q_7''(\tau)+2\pi v\cdot St\cdot q_7'(\tau) \tag{9a}$$

$$\beta_{2,7}''(\tau)-2\gamma\overline{\zeta}_2 v\cdot\left\{1-\frac{4f_{m,2}^2}{\hat{C}_{L2,v}^2}\left[e_{p7}\beta_{2,7}^2(\tau)+e_{p78}\beta_{2,8}^2(\tau)\right]\right\}\beta_{2,7}'(\tau)+v^2\cdot\beta_{2,7}(\tau)=\frac{D}{\overline{l}_2}q_7''(\tau)+2\pi v\cdot St\cdot q_7\prime(\tau)+\eta v\cdot\beta_{1,7}'(\tau) \tag{9b}$$

$$\beta_{1,8}''(\tau)-2\zeta_1 v\cdot\left\{1-\frac{4f_{m,1}^2}{\hat{C}_{L1}^2}\left[e_{p8}\beta_{1,8}^2(\tau)+e_{p87}\beta_{1,7}^2(\tau)\right]\right\}\beta_{1,8}'(\tau)+v^2\cdot\beta_{1,8}(\tau)=\frac{D}{\overline{l}_1}q_8''(\tau)+2\pi v\cdot St\cdot q_8'(\tau) \tag{9c}$$

$$\beta_{2,8}''(\tau)-2\gamma\overline{\zeta}_2 v\cdot\left\{1-\frac{4f_{m,2}^2}{\hat{C}_{L2,v}^2}\left[e_{p8}\beta_{2,8}^2(\tau)+e_{p87}\beta_{2,7}^2(\tau)\right]\right\}\beta_{2,8}'(\tau)+v^2\cdot\beta_{2,8}(\tau)=\frac{D}{\overline{l}_2}q_8''(\tau)+2\pi v\cdot St\cdot q_8\prime(\tau)+\eta v\cdot\beta_{1,8}'(\tau) \tag{9d}$$

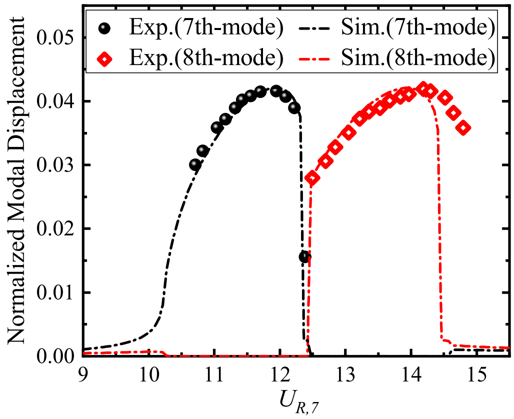


Fig. 7 Comparison of the simulated and measured multi-mode VIV response

$$q_7''(\tau) + \left[2\zeta_{s,7} + \frac{n_7 v}{2\pi St} \cdot \left(f_{m,1} + f_{m,2}\right)\right] q_7'(\tau) + q_7(\tau) = n_7 \cdot \left(\frac{v}{2\pi St}\right)^2 \cdot \left[f_{m,1} \cdot \beta_{1,7}(\tau) + f_{m,2} \cdot \beta_{2,7}(\tau)\right] \tag{9e}$$

$$q_8''(\tau) + \left[2\zeta_{s,8} \cdot \mu_8 + \frac{n_8 v}{2\pi St} \cdot \left(f_{m,1} + f_{m,2}\right)\right] q_8'(\tau) + \mu_8^2 \cdot q_8(\tau) = n_8 \cdot \left(\frac{v}{2\pi St}\right)^2 \cdot \left[f_{m,1} \cdot \beta_{1,8}(\tau) + f_{m,2} \cdot \beta_{2,8}(\tau)\right] \tag{9f}$$

where,

$$e_{p7} = \frac{\int_0^L \sin^4\left(\frac{7\pi}{L}z\right)\mathrm{d}z}{\int_0^L \sin^2\left(\frac{7\pi}{L}z\right)\mathrm{d}z},\ e_{p8} = \frac{\int_0^L \sin^4\left(\frac{8\pi}{L}z\right)\mathrm{d}z}{\int_0^L \sin^2\left(\frac{8\pi}{L}z\right)\mathrm{d}z},$$

$$e_{p78} = \frac{\int_0^L \sin^2\left(\frac{7\pi}{L}z\right)\sin^2\left(\frac{8\pi}{L}z\right)\mathrm{d}z}{\int_0^L \sin^2\left(\frac{7\pi}{L}z\right)\mathrm{d}z},$$

$$e_{p87} = \frac{\int_0^L \sin^2\left(\frac{8\pi}{L}z\right)\sin^2\left(\frac{7\pi}{L}z\right)\mathrm{d}z}{\int_0^L \sin^2\left(\frac{8\pi}{L}z\right)\mathrm{d}z},$$

$$v = \frac{\omega_v}{\omega_{s,7}} = U_{R,7} \cdot St,\ \mu_8 = \frac{f_{n,8}}{f_{n,7}},\ n_7 = \frac{\rho D^2 L}{2m_7},\ n_8 = \frac{\rho D^2 L}{2m_8},\ \tau = \omega_{s,7} \cdot t.$$

In the above derivation process, additional stiffness terms of the oscillators induced by the modal interaction are omitted in Eq. (9), as the natural frequencies of the oscillators are mainly decided by the *St* number of the target cylinder and the motion of cylinder. From Eq. (9), it can be found that the coupling between the two modes are induced by Eq. (9a-9d), which means the damping terms of aerodynamic effect are responsible for the VIV coupling between different modes.

4.2 Validation of the multi-mode VIV model

Using Eq. (9), the estimated VIV response amplitude of the target beam related to the 7th and 8th modes varying with wind speed can be calculated, as shown in Fig. 7. The response amplitudes that recorded in the experiment are also shown in Fig. 7. It is clear that, the estimated response by the mathematic model shows excellent agreement with the experiment, especially the mutual exclusion phenomenon between the two modes, indicating the feasibility of the proposed model for the analysis of multi-mode VIV. The difference of VIV wind speed range with simulation and experiment was discussed in Fig. 6. The miss of experimental data for $U_{R,7}$ = [10.0–10.5] is because that VIV of mode 6 suppresses the early stage of VIV of mode 7, as shown in Fig. 4. In case Mode 6 is also included into the model, such difference can be removed.

4.3 Effect of loading regime

Figure 8 shows the structure response amplitudes for the case of increasing wind speed, which follows the loading regime of the experiment. The case of decreasing wind speed is also examined. It is clear that, the response paths corresponding to the two regimes are only identical within some narrow ranges of wind speed. Although the responses different for the two cases, they still share one comment feature, that is no matter what loading regime is conducted, only one mode can be excited in one condition. Such mutual exclusive feature between different mode will be further discussed. This phenomenon is a typical feature for a nonlinear system, and it will be discussed in the following section.

5 Detailed analysis of response characteristics

The responses have been calculated in Sect. 4 using Runge–Kutta method, a time domain method. In that case, the unstable solutions could not be found, which means we could not have a full vision on the solutions of the system. Frequency domain method can, however, solve this problem.

In this section, a frequency domain method is adopted to calculate the response for detailed analysis. Incremental harmonic balance (IHB) method has been proved to have strong capacity on analyzing multi-DoF nonlinear dynamic system [59, 60]. More recently, the extended IHB (EIHB) method was further developed to enlarge application scope and efficiency [61, 62]. This provides a tool for more comprehensive investigation of behaviors of non-linear system.

$$\mathbf{K}_{nl}(\mathbf{q}_0, \mathbf{q}'_0) = \begin{bmatrix} e_{p7}\zeta_1 \frac{8f_{m,1}^2}{\hat{C}_{L1}^2}\beta_{1,7}\beta'_{1,7} & 0 & e_{p78}\zeta_1 \frac{8f_{m,1}^2}{\hat{C}_{L1}^2}\beta_{1,8}\beta'_{1,7} & 0 & 0 & 0 \\ 0 & e_{p7}\zeta_2 \frac{8f_{m,2}^2}{\hat{C}_{L2,v}^2}\beta_{2,7}\beta'_{1,7} & 0 & e_{p78}\zeta_2 \frac{8f_{m,2}^2}{\hat{C}_{L2,v}^2}\beta_{2,6}\beta'_{2,5} & 0 & 0 \\ e_{p87}\zeta_1 \frac{8f_{m,1}^2}{\hat{C}_{L1}^2}\beta_{1,7}\beta'_{1,8} & 0 & e_{p8}\zeta_1 \frac{8f_{m,1}^2}{\hat{C}_{L1}^2}\beta_{1,8}\beta'_{1,8} & 0 & 0 & 0 \\ 0 & e_{p87}\zeta_2 \frac{8f_{m,2}^2}{\hat{C}_{L2,v}^2}\beta_{2,7}\beta'_{2,8} & 0 & e_{p8}\zeta_2 \frac{8f_{m,2}^2}{\hat{C}_{L2,v}^2}\beta_{2,8}\beta'_{2,8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

5.1 Incremental harmonic balance method

According to Eq. (8), the governing functions are re-rewritten into matrix form,

$$\omega^2\mathbf{M}\mathbf{q}'' + \omega\mathbf{C}_{l1}\mathbf{q}' + \omega v\mathbf{C}_{l2}\mathbf{q}' + v\omega\mathbf{C}_{nl}\mathbf{q}' + v^2\mathbf{K}_1\mathbf{q} + \mathbf{K}_2\mathbf{q} = \mathbf{0}, \tag{10}$$

where, $\mathbf{M}$, $\mathbf{C}_{l1}$, $\mathbf{C}_{l2}$, $\mathbf{K}_1$, $\mathbf{K}_2$ are provided in "Appendix", ω is the ratio between vibration frequency and $\omega_{s,7}$.

Let ω_0, $\mathbf{q}_0$ and v_0 be one set of initial values of real system dynamic responses. A solution set close to this system state can be expressed as

$$\mathbf{q} = \mathbf{q}_0 + \Delta\mathbf{q}, \quad \omega = \omega_0 + \Delta\omega, \ v = v_0 + \Delta v, \tag{11}$$

Equation (11) is substituted into Eq. (10) to get

$$\begin{aligned} &\omega_0^2\mathbf{M}\Delta\mathbf{q}'' + \omega_0\left[\mathbf{C}_{l1} + v_0\mathbf{C}_{l2} + v_0\mathbf{C}_{nl}\left(\mathbf{q}_0^2\right)\right]\Delta\mathbf{q}' \\ &\quad + \left[v_0^2\mathbf{K}_1 + \mathbf{K}_2 + 2\omega_0\mathbf{K}_{nl}\left(\mathbf{q}_0, \mathbf{q}'_0\right)\right]\Delta\mathbf{q} \\ &\quad + \left\{2\omega_0\mathbf{M}\mathbf{q}''_0 + \left[\mathbf{C}_{l1} + v_0\mathbf{C}_{l2} + v_0\mathbf{C}_{nl}\left(\mathbf{q}_0^2\right)\right]\mathbf{q}'_0\right\} \\ &\quad \Delta\omega = \mathbf{R} - \left\{\omega_0\left[\mathbf{C}_{l2} + \mathbf{C}_{nl}\left(\mathbf{q}_0^2\right)\right]\mathbf{q}'_0 + 2v_0\mathbf{K}_1\mathbf{q}_0\right\}\Delta v, \end{aligned} \tag{12}$$

where

$$\mathbf{R} = -\left\{\omega_0^2\mathbf{M}\mathbf{q}''_0 + \omega_0\left[\mathbf{C}_{l1} + v_0\mathbf{C}_{l2} + v_0\mathbf{C}_{nl}\left(\mathbf{q}_0^2\right)\right]\mathbf{q}'_0 + \left(v_0^2\mathbf{K}_1 + \mathbf{K}_2\right)\mathbf{q}_0\right\}.$$

The response is assumed in the form of a Fourier series, and there are m components with different frequencies:

$$\mathbf{q}_{j0} = \sum_{k=1}^{m} a_{jk}\cos(k-1)\tau + \sum_{k=1}^{m-1} b_{jk}\sin(k\tau) = \mathbf{C}_s\mathbf{A}_j, \tag{13}$$

$$\Delta\mathbf{q}_{j0} = \sum_{k=1}^{m} \Delta a_{jk}\cos(k-1)\tau + \sum_{k=1}^{m-1} \Delta a_{jk}\sin(k\tau) = \mathbf{C}_s\Delta\mathbf{A}_j, \tag{14}$$

where Cs and A are the harmonic terms and corresponding amplitudes of the Fourier series of the response,

$$\mathbf{C}_s=[1\ \cos(\tau)\ \cos(2\tau)\ \cdots\ \cos((m-1)\tau)\ \sin(\tau)\ \sin(2\tau)\ \cdots\ \sin((m-1)\tau)],$$

$$\mathbf{A}_j=\begin{bmatrix} a_{j1} & a_{j2} & \cdots & a_{jm} & b_{j1} & b_{j2} & \cdots & b_{j(m-1)}\end{bmatrix}^{\mathrm{T}},\quad (j=1,2,\ldots,6)$$

$$\Delta\mathbf{A}_j=\begin{bmatrix} \Delta a_{j1} & \Delta a_{j2} & \cdots & \Delta a_{jm} & \Delta b_{j1} & \Delta b_{j2} & \cdots & \Delta b_{j(m-1)}\end{bmatrix}^{\mathrm{T}}\quad (j=1,2,\ldots,6)$$

The determination of m is flexible as it is related to the required accuracy of the solution. The larger this value the more accurate the solution will be. Therefore, there is no strict rule for the selection of this value. While, this number is basically related to the intense of nonlinearity. It should be larger if the target system possesses strong nonlinearity. In this study, $m = 5$ is adopted by examining the convergence of solutions.

Equations (13) and (14) can be rewritten as

$$\mathrm{q}_0=\mathrm{SA},\quad \Delta\mathrm{q}=\mathrm{S}\Delta\mathrm{A} \tag{15}$$

where $\mathbf{S} = \mathrm{diag}(\mathbf{C}_s, \mathbf{C}_s, \ldots, \mathbf{C}_s)$, $\mathbf{A}=\left[\mathbf{A}_1^{\mathrm{T}},\mathbf{A}_2^{\mathrm{T}},\ldots,\mathbf{A}_n^{\mathrm{T}}\right]^{\mathrm{T}}$. Equation (15) is then substituted into Eq. (12). Galerkin process is then applied to one cycle with τ between null to 2π such that the incremental linear equation as

$$\begin{bmatrix}\overline{\mathbf{U}} & \overline{\mathbf{K}}_{\mathrm{mc}}\end{bmatrix}\begin{bmatrix}\Delta\omega\\ \Delta\mathbf{A}\end{bmatrix}=\overline{\mathbf{R}}-\overline{\mathbf{R}}_{\mathrm{mc}}\Delta v, \tag{16}$$

where

$$\overline{\mathbf{K}}_{\mathrm{mc}}=\int_0^{2\pi}\mathbf{S}^{\mathrm{T}}\left\{\omega_0^2\mathbf{M}\mathbf{S}''+\omega_0\left[\mathbf{C}_{l1}+v_0\mathbf{C}_{l2}+v_0\mathbf{C}_{nl}\left(\mathbf{q}_0^2\right)\right]\mathbf{S}'+\left[v_0^2\mathbf{K}_1+\mathbf{K}_2+2\omega_0\mathbf{K}_{nl}\left(\mathbf{q}_0,\mathbf{q}'_0\right)\right]\mathbf{S}\right\}\mathrm{d}\tau,$$

$$\overline{\mathbf{U}}=\int_0^{2\pi}\mathbf{S}^{\mathrm{T}}\left\{2\omega_0\mathbf{M}\mathbf{S}''+\left[\mathbf{C}_{l1}+v_0\mathbf{C}_{l2}+v_0\mathbf{C}_{nl}\left(\mathbf{q}_0^2\right)\right]\mathbf{S}'\right\}\mathrm{d}\tau\cdot\mathbf{A},$$

$$\overline{\mathbf{R}}=-\int_0^{2\pi}\mathbf{S}^{\mathrm{T}}\left\{\omega_0^2\mathbf{M}\mathbf{S}''+\omega_0\left[\mathbf{C}_{l1}+v_0\mathbf{C}_{l2}+v_0\mathbf{C}_{nl}\left(\mathbf{q}_0^2\right)\right]\mathbf{S}'+\left(v_0^2\mathbf{K}_1+\mathbf{K}_2\right)\mathbf{S}\right\}\mathrm{d}\tau\cdot\mathbf{A},$$

$$\overline{\mathbf{R}}_{\mathrm{mc}}=\int_0^{2\pi}\mathbf{S}^{\mathrm{T}}\left\{\omega_0\left[\mathbf{C}_{l2}+\mathbf{C}_{nl}\left(\mathbf{q}_0^2\right)\right]\mathbf{S}'+2v_0\mathbf{K}_1\mathbf{S}\right\}\mathrm{d}\tau\cdot\mathbf{A}.$$

Equation (16) can then be solved numerically with the arc length method, and the responses are obtained iteratively for each increments of $[\Delta\omega\ \Delta\mathbf{A}^T]^T$ and Δv.

5.2 Results and discussion

By setting $\frac{D}{l_1} = 2.0$, $\frac{D}{l_2} = 1/1.9$, $f_{m,1} = 2.8$, $f_{m,2} = 18.0$, $\eta = 0.2$, $\hat{C}_{L1} = 0.27$, $\hat{C}_{L2,v} = 0.125$, $\zeta_1 = 0.2$, $\zeta_2 = 0.18$, $\zeta_7 = \zeta_8 = 0.6\%$, $St = 0.091$, $n_7 = n_8 = 0.00032$, with slight changes from those identified from the experiment, the response can be obtained as shown in Fig. 9. The solid line in Fig. 9 indicates the stable solution, and the dashed line means unstable solution. The stability of solution is examined using Floquet theory [56]. Unstable solution means it could not be observed in any situation in physical world, and unstable solution provides a boundary for the system's initial condition, which leads to different stable solutions. It can be checked from Fig. 9 that two solution branches can be obtained from the above system. For each branch, there is only one mode can be excited. That means, if Mode 7 is excited as shown in Fig. 9a, then Mode 8 will no longer vibrate, and vice versa (Fig. 9b).

Besides the mutual exclusive feature between different modes, one can also find the saddle node bifurcations from Fig. 9 for both modes. That means there are two stable solutions for the studied branch in between the two bifurcation points, $U_{R,7} \in [12.4, 13.9]$ for branch 1, and $U_{R,7} \in [14.3, 16.0]$ for branch 2. In this case, which solution the system converging to is determined by the initial condition of the system. Besides, it should be noted for this system, when $U_{R,7} \in [12.4, 13.9]$, as Mode 8 may also be excited in this wind speed range. Therefore, it can be known that, there are at least three sets of solutions the system may finally be converged to.

Figure 10 shows the vibration frequency of the system (including both the oscillators and structure). It can be clearly checked that when the 7th mode is excited, there is a lock-in range for $U_{R,7}$ from about 10.1 to 14.0. Similarly, when the 8th mode is excited, the lock-in range becomes 11.8–16.0. The blue pointed line in Fig. 10 indicates the Strouhal frequency ($St \times U_R$) of the vortex shedding varying with wind speed. The lock-in phenomenon also well simulates the physical phenomenon observed in VIV, which again suggests the feasibility of the proposed model.

6 Discussion of the mutual exclusion phenomenon

In Secs. 4 and 5, it is shown that the mutual exclusive VIV phenomenon between adjacent modes of bridge can be well simulated by the proposed model. This section will try to explain the such an interesting result from the mathematic point of view, based on the proposed model.

According to Eq. (2), it can be checked that the linear structure is excited by the two self-excited Van Der Pol (VDP) oscillators. Thus, the VDP oscillators are responsible for the nonlinear characteristics. The standard form of VDP oscillator function is as follows:

$$\ddot{x}(t) - \varepsilon\left[1 - x^2(t)\right]\dot{x}(t) + x(t) = 0, \tag{17}$$

where, x is the position coordinate, and ε is scalar parameter of the damping, which should be much lower than unity. It can be known from the equation

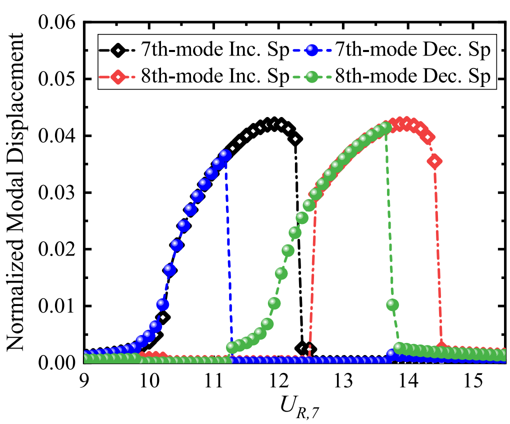


Fig. 8 Structure responses obtained from two loading regimes

that, on one hand, when the oscillation amplitude is small, the system has a negative damping, which leads to a gradual increase of oscillation amplitude; on the other hand, when the oscillation amplitude exceeds certain values, the system has a positive damping, which consequently results in a reduction of oscillation amplitude. Thus, the system will converge to a limit cycle oscillation (LCO).

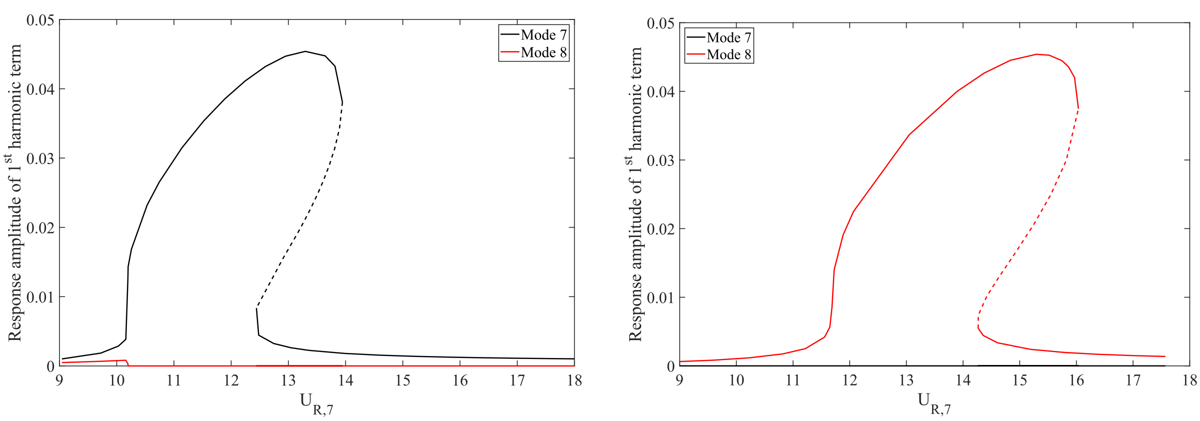
In case of multi-mode VIV, as illustrated in Eq. (9), it can be checked that the interaction between different modes is basically attributed to the coupled damping terms of VDP oscillators, as shown in Eq. (9a-9d). To make the interaction between the oscillators of two modes be more clearly illustrated, a coupling 2-dof VDP system is proposed following the format shown in Eq. (9):

$$\ddot{x}(t) - \varepsilon_1\left[1 - x^2(t) - y^2(t)\right]\dot{x}(t) + x(t) = 0, \tag{18a}$$

$$\ddot{y}(t) - \varepsilon_2\left[1 - y^2(t) - x^2(t)\right]\dot{y}(t) + \mu y(t) = 0, \tag{18b}$$

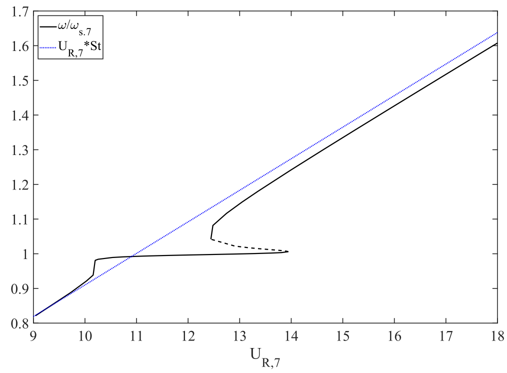
where, μ is a positive value close but not equal to unity, indicating the difference between the stiffness of the two oscillators, ε_1 and ε_2 controls nonlinear damping. In accordance to the discussion for Eq. (17), the system can be self-excited when the oscillation amplitude is small. As the damping of the two oscillators is coupled with each other as shown in Eq. (18), once one oscillator is strongly excited, it can increase the damping of the other oscillator at the same time, which results in the suppression of the oscillation of that oscillator.

Such feature will be illustrated by the following two numerical examples, by setting $\varepsilon_1 = 0.1$, $\varepsilon_2 = 0.11$, $\mu = 1.1$. Two different initial conditions $[x_0, y_0, \dot{x}_0, \dot{y}_0]$ = [0, 0, 0.1, 0.05] (Example 1) and [0, 0, 0.05, 0.1]

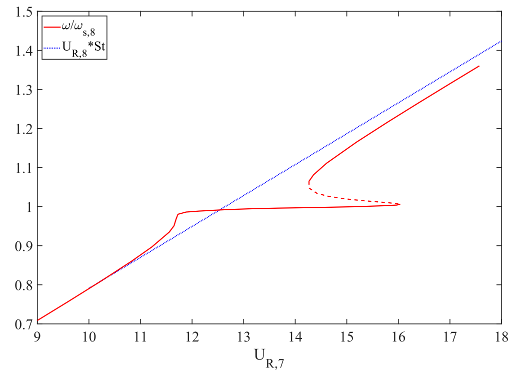


(a) Branch 1 (b) Branch 2

Fig. 9 Structure response obtained by IHB method



(a) Example 1



(b) Example 2

Fig. 10 Lock-in of the two modes

(Example 2) are adopted. Figure 11 shows the time series of the two oscillators starting from the two different initial conditions, respectively. It is clear that only one oscillator can finally be excited in the steady state for both examples. Which oscillator will finally be excited is determined by the initial condition of the system. In other words, the oscillator, who has the stronger initial oscillation, can gradually suppress the other one, as explained in the above paragraph.

According to Eq. (9), each mode of the beam is excited by two VDP oscillators, with similar format as shown in Eq. (18). That means only when the oscillators vibrate, the corresponding beam mode can be excited, and vise versa. Thus, such result can well explain the reason of the mutual exclusive phenomenon between adjacent modes.

7 Conclusions

In this study, VIV of a multi-span elastically supported beam is tested in wind tunnel experiment to examine the basic aeroelastic feature of long span bridge. The multi-mode VIV is observed in the experiment, and the mutual exclusive feature of VIV between different modes is checked. To further mathematically explain the phenomena, a dual-oscillator model is extended to simulate the multi-mode VIV of the long span bridge with 6:1 rectangular section. Following conclusions are obtained from this study:

(1) The dual-oscillator model can well simulate the whole VIV process of any specific mode of the bridge, by using only one set of model parameters. The results indicate that this model is feasible for simulating the VIV of 6:1 rectangular section cylinder, for the case with low turbulence intensity, including both the response amplitude and range of VIV.

(2) In order to further investigate the multi-mode VIV of long span bridge, the single mode model is extended to a multi-mode model, through considering to adjacent modes. By calculating the response of the multi-mode model, it is found that the proposed model can well simulate the entire VIV process for the two target modes, especially the mutual exclusive feature of VIV between adjacent modes can also be well captured by the proposed model.

(3) Incremental harmonic balance method is adopted to solve the governing functions in frequency domain, for further investigation of the system response. Two branches of the system solution are obtained from the system, and for each branch only one mode can be excited, which again illustrate the special feature of the system. Saddle node bifurcations can be checked from the solution, which is also a well-known characteristics of VIV. Moreover, the lock-in ranges of the investigated modes are identified from the obtained results.

(4) With further exploring the proposed model, it is found that the coupled Van Der Pol functions are responsible for the mutual exclusion of the VIV to two adjacent modes, as the large vibration of one mode can result in large damping of the other mode, which can cause the suppression of the vibration of that mode.

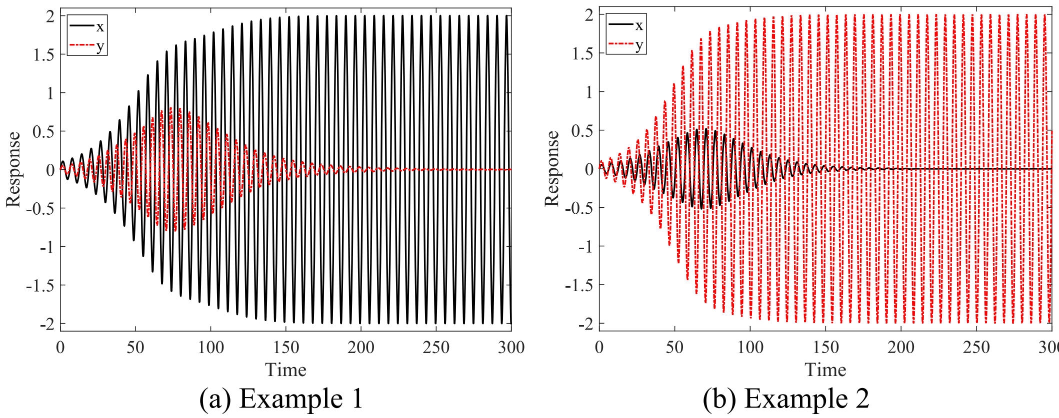


Fig. 11 Response of coupled VDP oscillators

> Based on the model, it can be known that, in real world when wind speed is just located at the overlapped VIV range of two adjacent modes, the initial condition of the bridge can determine which mode can be finally excited.

All the results obtained in this study suggest that the proposed model is a feasible tool for investigating the multi-mode VIV of long span bridge. Limitation of this study is that although the mutual exclusive phenomenon can be well simulated using mathematic model proposed in this study, the physical mechanism such phenomenon has not been well investigated. CFD simulation may still be needed for investigating the multi-mode VIV.

Acknowledgements The authors are grateful for the financial supports by the National Key Research and Development Program of China (2023YFC3805202), Natural Science Foundation of China (52221002), the 111 project of the Ministry of Education and the Bureau of Foreign Experts of China (No. B18062), the Natural Science Foundation of Chongqing, China (cstc2024ycjh-bgzxm0021).

Author contributions Yi Hui: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Validation. Xugang Hua: Experiment. Yuanyan Tang: Modelling, Writing – review. Weidong Zhu: Methodology. Tianyou Tao: Writing – review.

Funding The authors are grateful for the financial supports by the National Key Research and Development Program of China (2023YFC3805202), Natural Science Foundation of China (52221002), the 111 project of the Ministry of Education and the Bureau of Foreign Experts of China (No. B18062), the Natural Science Foundation of Chongqing, China (cstc2024ycjh-bgzxm0021).

Data availability All data, model, or code that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

$$\mathbf{M} = \begin{bmatrix} 1 & 0 & 0 & 0 & -\frac{D}{\bar{l}_1} & 0 \\ 0 & 1 & 0 & 0 & -\frac{D}{\bar{l}_2} & 0 \\ 0 & 0 & 1 & 0 & 0 & -\frac{D}{\bar{l}_1} \\ 0 & 0 & 0 & 1 & 0 & -\frac{D}{\bar{l}_2} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

$$\mathbf{K}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu_6^2 \end{bmatrix},$$

$$\mathbf{K}_1 = v^2 \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ -f_{m,1} n_5 \left(\frac{1}{2\pi St}\right)^2 & -f_{m,2} n_5 \left(\frac{1}{2\pi St}\right)^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & -f_{m,1} n_6 \left(\frac{1}{2\pi St}\right)^2 & -f_{m,2} n_6 \left(\frac{1}{2\pi St}\right)^2 & 0 & 0 \end{bmatrix},$$

$$\mathbf{C}_{l1} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2\zeta_{s,5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 2\zeta_{s,6}\mu_6 \end{bmatrix},$$

$$\mathbf{C}_{l2} = v \begin{bmatrix} -2\zeta_1 & 0 & 0 & 0 & -2\pi St & 0 \\ -\eta & -2\zeta_2 & 0 & 0 & -2\pi St & 0 \\ 0 & 0 & -2\zeta_1 & 0 & 0 & -2\pi St \\ 0 & 0 & -\eta & -2\zeta_2 & 0 & -2\pi St \\ 0 & 0 & 0 & 0 & \frac{n_5}{2\pi St} \cdot \left(f_{m,1} + f_{m,2}\right) & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{n_6}{2\pi St} \cdot \left(f_{m,1} + f_{m,2}\right) \end{bmatrix},$$

$$\mathbf{C}_{nl}(\mathbf{q}^2) = v \begin{bmatrix} \zeta_1 \frac{8f_{m,1}^2}{\hat{C}_{L1}^2} P_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \zeta_2 \frac{8f_{m,2}^2}{\hat{C}_{L2,v}^2} P_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \zeta_1 \frac{8f_{m,1}^2}{\hat{C}_{L1}^2} P_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & \zeta_2 \frac{8f_{m,2}^2}{\hat{C}_{L2,v}^2} P_4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

where, $\zeta_2 = \overline{\zeta}_2 \gamma \ \ \omega_{s,5}, P_1 = e_{p5}\beta_{1,5}^2(\tau) + e_{p56}\beta_{1,6}^2(\tau)$, $P_2 = e_{p5}\beta_{2,5}^2(\tau) + e_{p56}\beta_{2,6}^2(\tau)$,-
, $P_4 = e_{p6}\beta_{2,6}^2(\tau) + e_{p65}\beta_{2,5}^2(\tau)$.

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